WEEK 13 · Nowcasting

GDP Nowcasting through a Bank of Korea Paper

Estimating the unreleased current quarter from the information released so far

Publication lag · Vintage · Ragged edge · DFM · Kalman update · News · DFM–LSTM

Key question — how do we estimate the GDP growth of the current quarter, not yet released,
using only the data actually released as of today?

Nowcasting is estimation that respects the information cutoff

Three messages running through this lecture

1

Data have two times: the reference month and the release date

Open only data available today; a blank may be not-yet-released, not zero activity.

2

DFM and the Kalman filter read the common movement of many indicators

Update the factors with released indicators only; predict months not yet arrived with the transition equation.

3

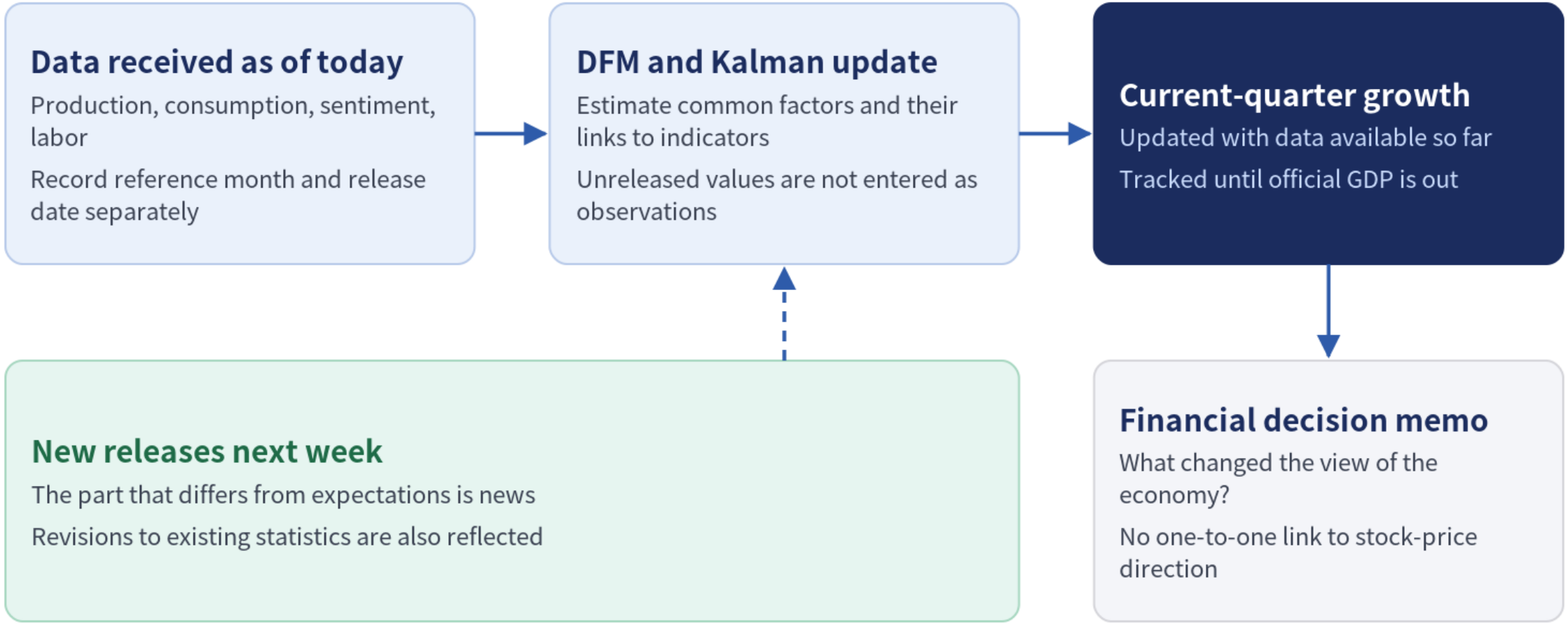
News is the gap from expectation, not the size of the released figure

Even high growth is negative news if the model expected more.

An application of Week 10's Kalman filter and a bridge to Week 15's deep learning comparison

Each week's new releases update the current-quarter forecast

Data actually received → DFM · Kalman update → current-quarter growth → decision memo



The reference paper is a system combining DFM and LSTM

Yi, Choi & Kim (2022), BOK Economic Analysis 28(2), 1–37

Location in paper	Content
pp. 7–11	DFM estimated in state-space form · Kalman filtering, smoothing, forecasting · parameters by EM
pp. 9–10	Four factor blocks: global, real, soft, labor · 5 quarterly and 29 monthly series
p. 10	DFM uses a 15-year rolling sample
pp. 7, 17–21	DFM monthly GDP representation and gap-filled data fed to LSTM · 15-month lookback · 500 model forecasts summarized
p. 7, fn. 8	Pre-2019 data are pseudo-vintages — not a fully real-time vintage over the whole period
pp. 22–25	Evaluation 2017Q1–2021Q3 · captured the crisis and rebound, but errors remain in the size of the plunge
p. 24, Fig. 9	The LSTM band is quantiles of 500 forecasts — not a definitive prediction interval

Check three places in the original yourself

Do not generalize the paper's results into universal performance today

p. 10 data setup · p. 19 Figure 7 DFM–LSTM flow · p. 24 Figure 9 forecast path and band

Questions to ask while reading

- Which data were used, and as of what date?
- What does the DFM do, and what does the LSTM do?
- What uncertainty does the band represent?

Source and scope

- DOI 10.23299/bokeri.2022.28.2.001 · Supplement: BOK Issue Note No. 2022-7
- A 2022 study — do not assume it equals the current operational system

01

PART 1 · Data

Publication lag · mixed frequency · vintages

Defining the nowcast · reference month and release date · Ragged edge

A nowcast estimates the current quarter in progress

Economic activity has already happened, but statistics are released later



Today is mid-quarter — GDP for all three quarters may still be unreleased

Present and available information

The present is the forecast target's reference time; available information is the data actually released by today

Name boundaries

Practical systems track the same quarter until the advance estimate is released

Open data by release date, not by reference month

Hypothetical release dates — not actual Korean statistics release dates

Indicator	Reference month	Assumed release date	Usable on Apr 10
Production	Feb	Mar 31	Yes
Production	Mar	Apr 30	No
Sentiment	Mar	Mar 25	Yes
Q1 GDP	Jan–Mar	Apr 25	No

Common leak

Aligning by reference month only and including March values lets future information in

Recording rule

If the release time matters, record the time as well as the date

A vintage is a snapshot of the data held on that day

In reports, distinguish true real-time replication from experiments that only mimic publication lags

Type	Information held	Limitation
True vintage	Values as released and revised at the time	Requires data archiving
Pseudo-vintage	Publication lags applied to latest values	Cannot restore past revisions exactly
Latest complete panel	Final values known today	Leaks if used for past real-time evaluation

In the reference paper

- Pseudo-vintages were used before 2019 — do not read the whole period as fully real-time

A blank at the data edge is unreleased, not zero

Hypothetical April 10 vintage — each indicator ends in a different month · the quarterly GDP row is conceptual

	Dec	Jan	Feb	Mar
Production	Available	Available	Available	Unreleased
Consumption	Available	Available	Available	Unreleased
Sentiment	Available	Available	Available	Available
Labor	Available	Available	Available	Unreleased
Quarterly GDP	Available	Unreleased	Unreleased	Unreleased

Do not insert 0

A blank is not zero activity but not yet released

Do not drop rows

Dropping all recent rows to match the slowest series loses information

Monthly indicators and quarterly GDP run on different clocks

Putting mixed-frequency data into one model

Simply averaging monthly variables to quarters is convenient but misses the latest month's information

Mixed-frequency DFM

- Use time aggregation and a state vector that match the data definitions
- The mixed-frequency setup in the paper's Appendix 2 is optional reading

Caution

- Do not generalize quarterly growth as 'the simple average of three monthly growth rates'

02

PART 2 · Models

DFM · Kalman · News · LSTM

Common factors · information cutoff · news decomposition · division of roles

A DFM summarizes many indicators with a few common factors

Minimal introductory equations — not the paper's full model

$$\underbrace{z_t = \Lambda f_t + \varepsilon_t}_{\text{observation equation}} \qquad \underbrace{f_t = A f_{t-1} + u_t}_{\text{state equation}}$$

Notation

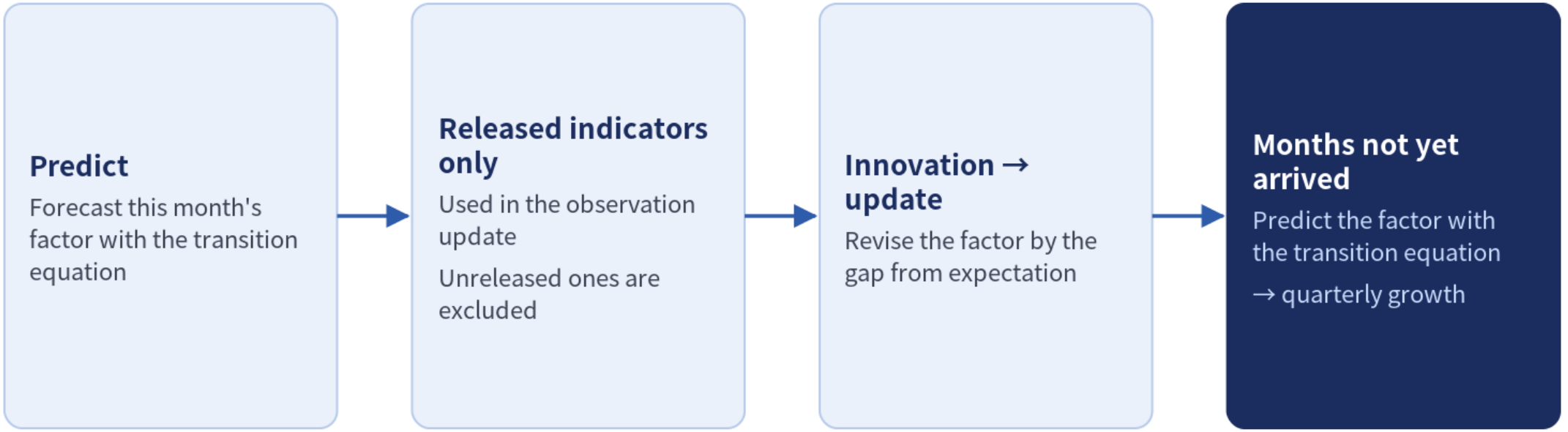
- Observed indicators z, common factors f
- Indicator loadings Λ, factor persistence A
- Lag structure of idiosyncratic errors goes into the state

Limits of interpretation

- Calling a factor 'the business cycle' does not mean the true state of the economy or causal identification

Apply Week 10's Kalman cycle to many economic indicators

Leave unreleased indicators out of the observation update; revise factors with released ones only

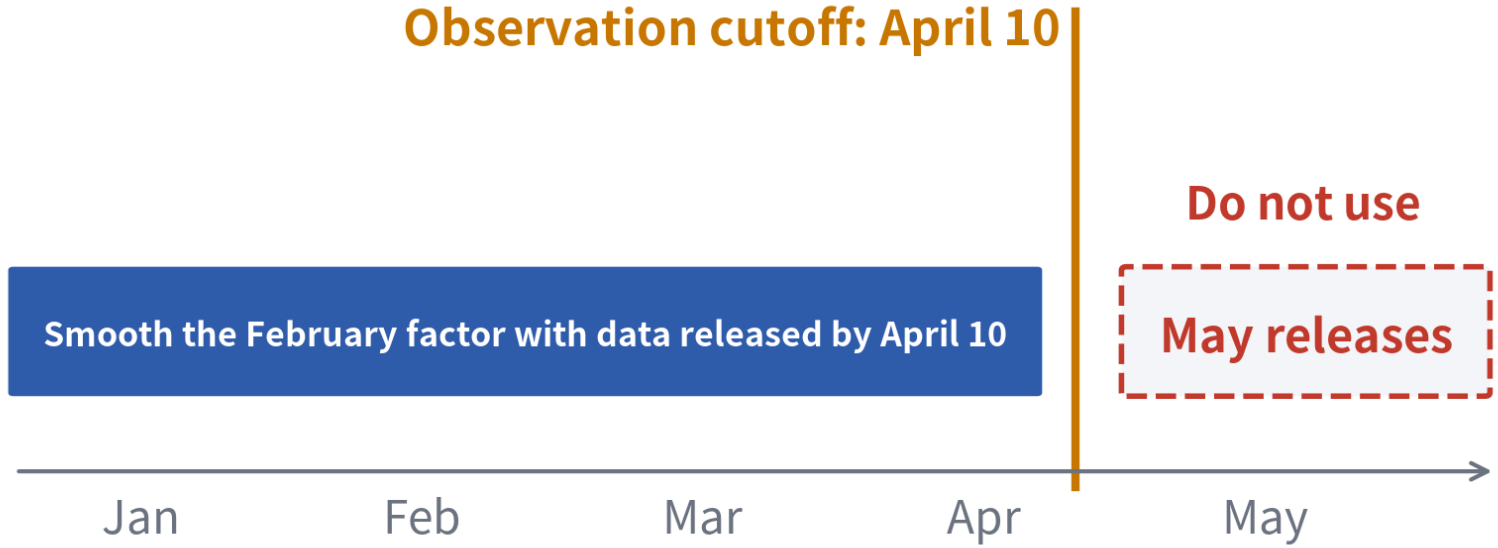


Linking months and quarters

Build the observation equation linking monthly factors to quarterly GDP to fit the economic definitions

Smoothing last month's factor with today's data is allowed

The rule is not 'no smoothing' but 'no observations after the decision date'



Allowed

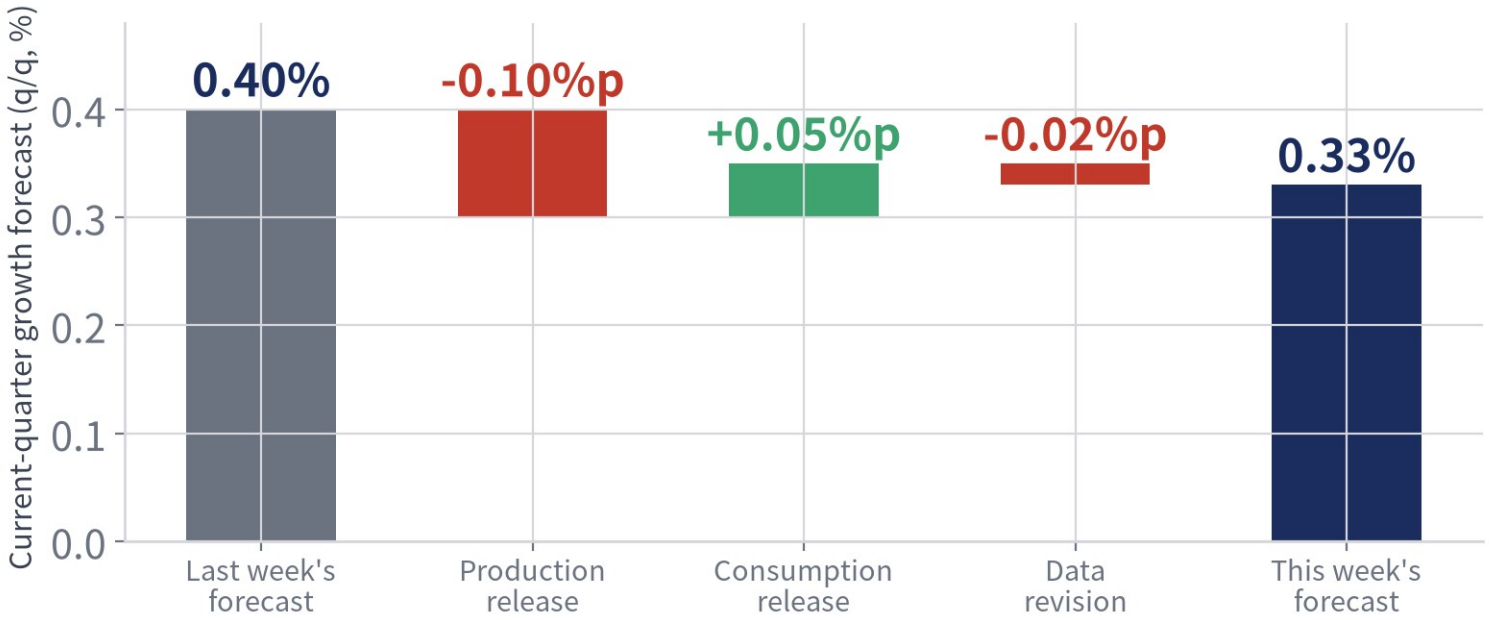
Re-estimating the February factor with data as of April 10

Not allowed

Adding May releases and recording it as the April 10 forecast

News is the gap from expectation, not the released figure

Hypothetical class example — not a result of the paper · weekly revisions of the q/q growth forecast



What sets the impact

Relevance to target GDP · uncertainty of the news · overlap with other news

Caution on summing

Do not simply add them assuming independence; show re-estimation effects separately

The update from one new release is set by a covariance ratio

Basic intuition under joint Gaussianity · the information set is pre-release information

$$\underbrace{\Delta \hat{g}}_{\text{forecast revision}} = \frac{\mathrm{Cov}(g, e \mid \mathcal{I})}{\mathrm{Var}(e \mid \mathcal{I})} \underbrace{e}_{\text{News: released value − prior expectation}}$$

How to read

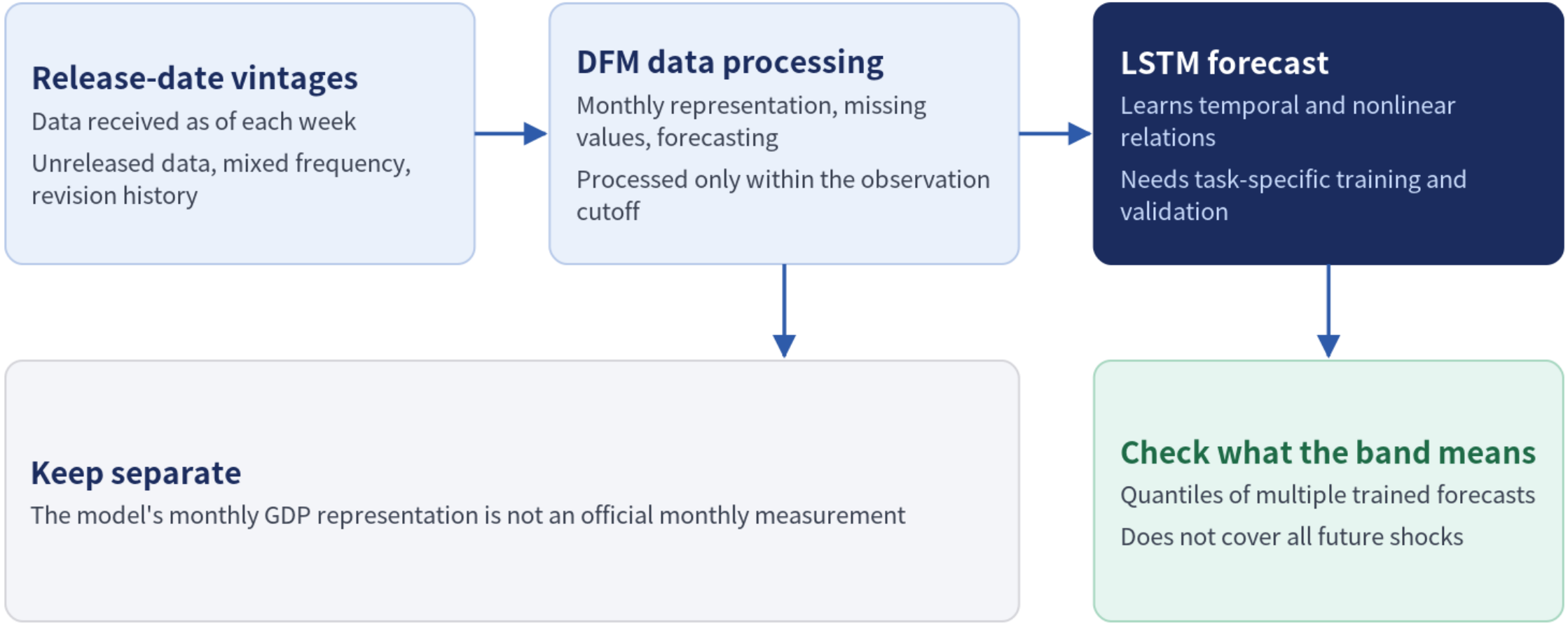
- The more the news co-moves with GDP and the less noisy it is, the more the forecast changes

Limitation

- Real data revisions and multi-release decompositions are more complex

The DFM structures the data and the LSTM learns the patterns

A simple average of two forecasts and passing DFM-processed inputs to an LSTM are different systems



An LSTM is a sequential network that learns to remember and forget

Internal structure and foundation-model comparisons return in Week 15

Raising input frequency does not automatically increase the number of independent quarterly GDP targets

Role

- A candidate for learning temporal patterns and nonlinear relations in the inputs

Sample dependence

- Overlapping monthly windows make training samples dependent

First ask what uncertainty the band in the figure shows

'500 models' are not '500 actual economies'

Source of band	What it shows	What it does not automatically include
LSTM forecasts from many initializations	Forecast spread from training randomness	All future shocks to GDP itself
State-space conditional interval	State uncertainty under the given model and parameters	All parameter and model errors
Forecast-error-based calibrated interval	Empirical range under validation conditions	The same coverage when the distribution shifts

Evaluation

Evaluate empirical coverage and interval performance in crises separately

03

PART 3 · Practice

Weekly nowcast updates

Replication level · practice · link to financial decisions

Name the work by its replication level

Unless it is level C, do not claim a full replication of the BOK paper

Level	Task	Correct name
A Required	Replicate the structure with synthetic indicators, release dates, hidden factors, quarterly target	Educational simulation
B Optional	Apply publication lags to public Korean economic data	Quasi-real-time · pseudo-vintage experiment
C Advanced	Restore first releases, revisions, and actual release logs	Vintage-based real-time evaluation

Scale

Training 34 variables and 500 LSTMs as in the paper is not required

Reproduce the paper's idea with a scaled-down simulation ①

Paste into Claude Code · data generation and vintage construction

[Input] Prompt · Claude Code

```
Build an educational nowcasting exercise that explains the idea
of the 2022 Bank of Korea DFM–LSTM paper.
State that this is a scaled-down simulation, not an exact replication.
1. With seed=42, generate 10 years of monthly synthetic data
   and a quarterly target. Build 4 indicators linked to a common
   factor: production, consumption, sentiment, labor. Give each
   a different release lag and create an actual release-date field.
2. Create a hidden monthly target level, aggregate it to quarterly
   levels, and compute growth. Explain the quarterly aggregation rule
   and do not blindly average monthly growth rates.
3. Open only values released by each as-of date; keep the rest missing.
   Do not input future-quarter GDP or revised values.
4. Implement a small mixed-frequency DFM or an explicitly simplified
   state-space model.
   Do not run PCA on arbitrary full-sample interpolation
   and call it an exact DFM.
```

What to check

- Were no values opened before release?
- Is the target unit q/q growth?
- Is it a true vintage or a pseudo-vintage?

Reproduce the paper's idea with a scaled-down simulation ②

Paste next · weekly tracking, baselines, evaluation

[Input] Prompt · Claude Code

```
5. Track one quarter on weekly as-of dates and store the forecasts.
   Separate release-news effects with fixed model parameters
   from re-estimation effects.
6. Compare against baselines: persistence of past growth and
   a simple bridge equation.
7. If possible, add a small LSTM, but
   split train/validation/test in time order.
   DFM input generation must also respect each information cutoff;
   do not train on test values.
8. Compare errors by time remaining until the release
   for the same quarter. Do not treat multiple weekly forecasts
   as independent GDP samples.
9. Explain with PNG figures, the availability matrix, news bars,
   and forecast vs. actual rather than equations.
```

What to check

- Did the target release leak into the inputs?
- Did retraining and all preprocessing respect the cutoff?
- Outputs: vintage matrix · weekly forecasts · release contributions · error evaluation

A nowcast becomes material for an investment memo

Assignment: a 1-page memo on 'Which release this week changed the view of the economy?'

Growth forecast

This week's current-quarter forecast and change from last week

Cause of revision

Which release differed from expectations · re-estimation effects separately

Uncertainty

Type and size of uncertainty the band represents

Gap vs. market expectations

Expectations already priced in · indicators to watch next

Higher forecast ≠ signal of higher rates or stocks — inflation, policy response, and market expectations may differ

Check understanding and pass the questions to Week 15

The first three check this week; the last starts Week 15's deep learning comparison

Q1	Can high production growth be negative news? — Yes, if the prior expectation was higher	**News**
Q2	Is smoothing last month's factor with today's data always a leak? — Not if no post-today data are used	**Information cutoff**
Q3	Is the band from many LSTM initializations a guaranteed interval for true GDP? — No	**Uncertainty**
Q4	Compare Chronos·TimesFM with the same release dates and inputs? Does pretraining help when targets are few?	**Week 15 link**

Estimating the present means respecting the time of the data

Next week: when even the parameters are uncertain

1

Reference month · release date · vintage

Open only data available today, and distinguish pseudo-vintages from true vintages.

2

DFM + Kalman = real-time updating of common factors

Revise factors with released indicators and predict unreleased months with the transition equation.

3

Read news and bands correctly

News is the gap from expectation; for a band, first check what kind of uncertainty it represents.

Do not assume foundation models automatically handle mixed frequency, missing values, and revision history — Week 15